

Foundations of Stochastic Calculus for Option Pricing

Itô's Lemma, the Girsanov Change of Measure, and the Feynman–Kac Representation

Mohit Kumar Jassi

Abstract

These notes give a from-first-principles derivation of the three pillars connecting the probabilistic and PDE views of derivative pricing: Itô's Lemma, the Girsanov change-of-measure theorem underlying risk-neutral pricing, and the Feynman–Kac representation linking conditional expectations to parabolic PDEs. Every step is justified rather than asserted; where a naive ordinary-calculus argument fails, the failure is made explicit before the correct stochastic argument is given.

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1 Itô Processes and Itô's Lemma

1.1 Itô processes

Definition 1.1 (Itô process). A process $(X_t)_{t \geq 0}$ generalises Brownian motion by allowing both a drift and a diffusion term. X_t is an *Itô process* if it can be written as

$$X_t = X_0 + \int_0^t \mu_s ds + \int_0^t \sigma_s dW_s,$$

or, in differential (SDE) notation,

$$dX_t = \underbrace{\mu_t}_{\text{drift}} dt + \underbrace{\sigma_t}_{\text{diffusion}} dW_t, \quad W_t \text{ a Wiener process.}$$

Remark 1.2 (Integrability conditions). Both μ_t and σ_t are required to be $\mathcal{F}_t$ -adapted and to satisfy suitable integrability conditions so that the two integrals above are well defined:

$$\int_0^t |\mu_s| ds < \infty \quad \text{a.s.}, \quad \int_0^t \sigma_s^2 ds < \infty \quad \text{a.s.}$$

The drift integral $\int_0^t \mu_s ds$ is an ordinary (pathwise) Riemann integral and pushes the process in a *predictable* direction. The stochastic integral $\int_0^t \sigma_s dW_s$ injects randomness whose variance accumulates over time, since $\text{Var}(dW_t) = dt$. We verify this accumulation explicitly:

$$\text{Var} \left[\int_0^t \sigma_s dW_s \right] = \mathbb{E} \left[\left(\int_0^t \sigma_s dW_s \right)^2 \right] - \underbrace{\left(\mathbb{E} \left[\int_0^t \sigma_s dW_s \right] \right)^2}_{=0} = \int_0^t \mathbb{E}[\sigma_s^2] ds,$$

using the Itô isometry $\mathbb{E} \left[\left(\int_0^t \sigma_s dW_s \right)^2 \right] = \mathbb{E} \left[\int_0^t \sigma_s^2 ds \right]$. For constant $\sigma_s \equiv \sigma$ this reduces to $\text{Var} \left[\int_0^t \sigma dW_s \right] = \sigma^2 t$, consistent with $X_t \sim \mathcal{N}(0, t)$ for standard Brownian motion. When σ_s varies with time or state, this variance must in general be recomputed at each instant — this state-dependence is precisely what Itô's Lemma below allows us to track exactly, rather than approximately.

1.2 Derivation of Itô's Lemma

We want to know how a smooth function $f(X_t, t)$ evolves when X_t is an Itô process. Ordinary calculus would Taylor-expand and retain first-order terms only; we show explicitly why this fails here.

Step 1: Multivariate Taylor expansion

For $f(x, t)$, expand around (X_t, t) with small increments dX_t, dt , keeping second-order terms:

$$f(X_t + dX_t, t + dt) = f(X_t, t) + f_t dt + f_x dX_t + \frac{1}{2} \left(f_{tt} (dt)^2 + 2f_{tx} dt dX_t + f_{xx} (dX_t)^2 \right) + \dots$$

Define $df := f(X_t + dX_t, t + dt) - f(X_t, t)$, so

$$df = f_t dt + f_x dX_t + \frac{1}{2} f_{tt} (dt)^2 + f_{tx} dt dX_t + \frac{1}{2} f_{xx} (dX_t)^2 + \dots$$

Step 2: Expand $(dX_t)^2$ and the mixed term

Substituting $dX_t = \mu dt + \sigma dW_t$:

$$dt \cdot dX_t = \mu (dt)^2 + \sigma dW_t dt, \tag{1}$$

$$(dX_t)^2 = \mu^2 (dt)^2 + 2\mu\sigma dt dW_t + \sigma^2 (dW_t)^2. \tag{2}$$

Step 3: Order analysis (the Itô multiplication table)

Using the heuristic $dW_t \sim \sqrt{dt}$ — justified since $\text{Var}(dW_t) = dt$, so the *typical size* of dW_t scales as $\sqrt{dt}$ — track the order in dt of every term:

$$(dt)^2 \rightarrow 0, \quad (dt)^{3/2} = dt \cdot dW_t \rightarrow 0, \quad (dW_t)^2 \rightarrow dt.$$

The first two vanish faster than dt itself and are discarded (this is the ordinary-calculus step). The last is the crux: $(dW_t)^2$ is the *same order* as dt , so it cannot be discarded — it is this term, and only this term, that survives from every second-order piece above. Consequently:

$$f_{tt}(dt)^2 \rightarrow 0, \quad f_{tx} dt dX_t \rightarrow 0, \quad f_{xx}(dX_t)^2 \rightarrow f_{xx} \sigma^2 dt.$$

Step 4: Collect terms

$$df = f_t dt + f_x(\mu dt + \sigma dW_t) + \frac{1}{2} f_{xx} \sigma^2 dt$$

Theorem 1.3 (Itô’s Lemma). *Let $dX_t = \mu_t dt + \sigma_t dW_t$ and let $f(x, t)$ be twice continuously differentiable in x and once in t . Then*

$$df = \left(f_t + \mu_t f_x + \frac{1}{2} \sigma_t^2 f_{xx} \right) dt + \sigma_t f_x dW_t$$

Remark 1.4. The term $\frac{1}{2} \sigma_t^2 f_{xx} dt$ — the *Itô correction* — is the entire difference from the chain rule of ordinary calculus. It exists because $(dW_t)^2 = dt$ rather than 0, which is itself a consequence of the nowhere-differentiable, infinite-quadratic-variation sample paths of Brownian motion.

1.3 Worked exercise: the GBM log-transform

Exercise. Let $f(S_t) = \ln S_t$, where $dS_t = \mu S_t dt + \sigma S_t dW_t$ (geometric Brownian motion). Find $d(\ln S_t)$.

Solution. Compute the partial derivatives of $f(x) = \ln x$:

$$f_t = 0, \quad f_x = \frac{1}{S_t}, \quad f_{xx} = -\frac{1}{S_t^2}.$$

Apply Theorem 1.3 with $\mu_t \rightarrow \mu S_t$, $\sigma_t \rightarrow \sigma S_t$:

$$d(\ln S_t) = \left(0 + \mu S_t \cdot \frac{1}{S_t} + \frac{1}{2} \sigma^2 S_t^2 \cdot \left(-\frac{1}{S_t^2} \right) \right) dt + \sigma S_t \cdot \frac{1}{S_t} dW_t$$

$$d(\ln S_t) = \left(\mu - \frac{1}{2} \sigma^2 \right) dt + \sigma dW_t$$

Integrating from 0 to T :

$$\int_0^T d(\ln S_t) = \left(\mu - \frac{\sigma^2}{2} \right) \int_0^T dt + \int_0^T \sigma dW_t \implies \ln \left(\frac{S_T}{S_0} \right) = \left(\mu - \frac{\sigma^2}{2} \right) T + \sigma W_T,$$

so that

$$S_T = S_0 \exp \left[\left(\mu - \frac{\sigma^2}{2} \right) T + \sigma W_T \right]$$

□

Remark 1.5. The $-\frac{1}{2} \sigma^2$ correction is why $\mathbb{E}[\ln S_T] \neq \ln \mathbb{E}[S_T]$: the log-price and the price level carry *different* drifts under the same dynamics — a fact with no ordinary-calculus analogue and a frequent source of interview errors.

2 The Radon–Nikodym Derivative and Girsanov’s Theorem

2.1 Motivation: risk-neutral drift

Under the real-world measure $\mathbb{P}$, a stock follows $dS_t = \mu S_t dt + \sigma S_t dW_t^{\mathbb{P}}$. For pricing we require a measure $\mathbb{Q}$ under which the drift is the risk-free rate r rather than μ . We construct $\mathbb{Q}$ by shifting the driving Brownian motion.

Definition 2.1 (Drift shift). For a process θ_t (the *market price of risk*), define

$$dW_t^{\mathbb{Q}} := dW_t^{\mathbb{P}} + \theta_t dt, \quad \text{equivalently} \quad W_t^{\mathbb{Q}} = W_t^{\mathbb{P}} + \int_0^t \theta_s ds.$$

Proposition 2.2 (Choice of θ_t). *The unique θ_t for which S_t has drift r under $\mathbb{Q}$ is $\theta_t = \frac{\mu - r}{\sigma}$ (the Sharpe ratio).*

Proof. Substitute $dW_t^{\mathbb{P}} = dW_t^{\mathbb{Q}} - \theta_t dt$ into the $\mathbb{P}$ -dynamics:

$$dS_t = \mu S_t dt + \sigma S_t (dW_t^{\mathbb{Q}} - \theta_t dt) = (\mu - \sigma \theta_t) S_t dt + \sigma S_t dW_t^{\mathbb{Q}}.$$

Comparing with the target $dS_t = r S_t dt + \sigma S_t dW_t^{\mathbb{Q}}$ requires $\mu - \sigma \theta_t = r$, i.e. $\theta_t = (\mu - r)/\sigma$. $\square$

What remains is to show that $W_t^{\mathbb{Q}}$, so defined, is *actually* a Brownian motion — under some genuine probability measure $\mathbb{Q}$. This is the content of Girsanov’s theorem, and requires first understanding the Radon–Nikodym machinery that makes a change of measure precise.

2.2 The Radon–Nikodym derivative

Definition 2.3 (Absolute continuity and RND). Let $\mathbb{P}, \mathbb{Q}$ be probability measures on $(\Omega, \mathcal{F})$. $\mathbb{Q}$ is *absolutely continuous* with respect to $\mathbb{P}$ (written $\mathbb{Q} \ll \mathbb{P}$) if, for every $A \in \mathcal{F}$, $\mathbb{P}(A) = 0 \implies \mathbb{Q}(A) = 0$: no measure can assign positive probability to an event the other calls impossible. If $\mathbb{Q} \ll \mathbb{P}$, the Radon–Nikodym theorem guarantees a random variable $Z = d\mathbb{Q}/d\mathbb{P}$ such that

$$\mathbb{Q}(A) = \int_A Z d\mathbb{P} = \mathbb{E}^{\mathbb{P}}[Z \mathbf{1}_A] \quad \forall A \in \mathcal{F}.$$

Remark 2.4 (Interpretation). Z is a path-by-path reweighting function. Where $Z > 1$, $\mathbb{Q}$ assigns more weight than $\mathbb{P}$; where $Z < 1$, less. The key consequence, used throughout risk-neutral pricing, is

$$\boxed{\mathbb{E}^{\mathbb{Q}}[X] = \mathbb{E}^{\mathbb{P}}[X \cdot Z]}$$

— an expectation under an unfamiliar measure $\mathbb{Q}$ is computed entirely under the familiar measure $\mathbb{P}$, after reweighting by Z .

Proposition 2.5 (Z is non-negative and has unit mean). *Any valid RND Z satisfies $Z \geq 0$ a.s. and $\mathbb{E}^{\mathbb{P}}[Z] = 1$.*

Proof. Since $\mathbb{Q}$ is a probability measure, $\mathbb{Q}(A) \geq 0$ for every $A \in \mathcal{F}$. If $Z < 0$ on some set B with $\mathbb{P}(B) > 0$, then $\mathbb{Q}(B) = \mathbb{E}^{\mathbb{P}}[Z \mathbf{1}_B] < 0$, a contradiction; hence $Z \geq 0$ a.s. Taking $A = \Omega$: $1 = \mathbb{Q}(\Omega) = \mathbb{E}^{\mathbb{P}}[Z \mathbf{1}_{\Omega}] = \mathbb{E}^{\mathbb{P}}[Z]$. $\square$

Remark 2.6 (Why Z must be *strictly* positive). Non-negativity alone does not require $Z > 0$ everywhere; that stronger requirement comes from *equivalence* of measures ($\mathbb{P} \sim \mathbb{Q}$, i.e. $\mathbb{Q} \ll \mathbb{P}$ and $\mathbb{P} \ll \mathbb{Q}$), which Girsanov’s construction needs so that the change of measure is invertible. If $Z_t = 0$ on some set, then $d\mathbb{P}/d\mathbb{Q} = 1/Z_t$ blows up there, meaning $\mathbb{P}$ assigns positive probability to an event $\mathbb{Q}$ considers impossible — breaking the two-way translation between $\mathbb{P}$ - and $\mathbb{Q}$ -expectations that risk-neutral pricing relies on constantly. As shown below, the exponential form of Z_t guarantees $Z_t > 0$ automatically, since $e^x > 0$ for all finite x .

2.3 Martingale property of the Radon–Nikodym process

Define the process $Z_t := \mathbb{E}^{\mathbb{P}}[Z_T \mid \mathcal{F}_t]$ for a fixed terminal $Z_T = d\mathbb{Q}/d\mathbb{P}$.

Proposition 2.7. $(Z_t)_{t \geq 0}$ is a $\mathbb{P}$ -martingale with $Z_0 = 1$.

Proof. By the tower property, for $s < t$,

$$\mathbb{E}^{\mathbb{P}}[Z_t \mid \mathcal{F}_s] = \mathbb{E}^{\mathbb{P}}[\mathbb{E}^{\mathbb{P}}[Z_T \mid \mathcal{F}_t] \mid \mathcal{F}_s] = \mathbb{E}^{\mathbb{P}}[Z_T \mid \mathcal{F}_s] = Z_s.$$

$Z_0 = \mathbb{E}^{\mathbb{P}}[Z_T] = 1$ by Proposition 2.5. □

Remark 2.8. This is structural — it requires no extra assumption. By the Martingale Representation Theorem, any $\mathbb{P}$ -martingale adapted to a Brownian filtration can be written as a pure stochastic integral against $dW_t^{\mathbb{P}}$ with zero drift:

$$dZ_t = \gamma_t dW_t^{\mathbb{P}}$$

for some (as yet unknown) process γ_t .

2.4 Deriving γ_t : Lévy’s characterization and the abstract Bayes rule

We now determine γ_t — and hence Z_t explicitly — by requiring that $W_t^{\mathbb{Q}}$, as defined above, genuinely be a $\mathbb{Q}$ -Brownian motion.

Theorem 2.9 (Lévy’s characterization of Brownian motion). *A continuous process (M_t) with $M_0 = 0$ is a Brownian motion under a measure $\mathbb{Q}$ if and only if it is a $\mathbb{Q}$ -martingale and its quadratic variation is $\langle M \rangle_t = t$.*

Remark 2.10. Since $W_t^{\mathbb{Q}} = W_t^{\mathbb{P}} + \int_0^t \theta_s ds$ differs from $W_t^{\mathbb{P}}$ only by a finite-variation drift term, quadratic variation is unaffected: $\langle W^{\mathbb{Q}} \rangle_t = \langle W^{\mathbb{P}} \rangle_t = t$. The entire content of Girsanov’s theorem therefore reduces to showing $W_t^{\mathbb{Q}}$ is a $\mathbb{Q}$ -martingale.

Lemma 2.11 (Abstract Bayes rule). *For equivalent measures with RND process Z_t ,*

$$\mathbb{E}^{\mathbb{Q}}[X_t \mid \mathcal{F}_s] = \frac{\mathbb{E}^{\mathbb{P}}[Z_t X_t \mid \mathcal{F}_s]}{Z_s}.$$

Consequently, X_t is a $\mathbb{Q}$ -martingale $\iff Z_t X_t$ is a $\mathbb{P}$ -martingale.

Applying this with $X_t = W_t^{\mathbb{Q}}$: $W_t^{\mathbb{Q}}$ is a $\mathbb{Q}$ -martingale iff $Z_t W_t^{\mathbb{Q}}$ is a $\mathbb{P}$ -martingale. This converts an unfamiliar $\mathbb{Q}$ -calculation into a $\mathbb{P}$ -calculation we can carry out directly with Itô’s Lemma.

Proposition 2.12 (Determining γ_t). $Z_t W_t^{\mathbb{Q}}$ is a $\mathbb{P}$ -martingale if and only if $\gamma_t = -Z_t \theta_t$.

Proof. Apply the Itô product rule to $Z_t W_t^{\mathbb{Q}}$, using $dW_t^{\mathbb{Q}} = dW_t^{\mathbb{P}} + \theta_t dt$ and $dZ_t = \gamma_t dW_t^{\mathbb{P}}$:

$$d(Z_t W_t^{\mathbb{Q}}) = Z_t dW_t^{\mathbb{Q}} + W_t^{\mathbb{Q}} dZ_t + d\langle Z, W^{\mathbb{Q}} \rangle_t.$$

Term by term:

$$Z_t dW_t^{\mathbb{Q}} = Z_t dW_t^{\mathbb{P}} + Z_t \theta_t dt,$$

$$W_t^{\mathbb{Q}} dZ_t = W_t^{\mathbb{Q}} \gamma_t dW_t^{\mathbb{P}},$$

$d\langle Z, W^{\mathbb{Q}} \rangle_t = \gamma_t dt$ (the $\theta_t dt$ piece of $W^{\mathbb{Q}}$ is finite-variation and contributes 0 to covariation).

Summing and separating drift from diffusion:

$$d(Z_t W_t^{\mathbb{Q}}) = \underbrace{(Z_t \theta_t + \gamma_t)}_{\text{drift}} dt + \underbrace{(Z_t + W_t^{\mathbb{Q}} \gamma_t)}_{\text{diffusion}} dW_t^{\mathbb{P}}.$$

For a martingale the drift must vanish: $Z_t \theta_t + \gamma_t = 0 \Rightarrow \gamma_t = -Z_t \theta_t$. This is the *unique* value of γ_t consistent with $W_t^{\mathbb{Q}}$ being a $\mathbb{Q}$ -Brownian motion. □

2.5 Solving for Z_t explicitly

We must now solve

$$dZ_t = -Z_t \theta_t dW_t^{\mathbb{P}}, \quad Z_0 = 1. \quad (*)$$

Remark 2.13 (Why the exponential ansatz, and why a drift term in the exponent). The SDE $(*)$ is *multiplicative* — the diffusion coefficient is Z_t itself, exactly as in GBM. A naive division suggests $d(\ln Z_t) = -\theta_t dW_t^{\mathbb{P}}$, hence $\ln Z_t = -\int_0^t \theta_s dW_s^{\mathbb{P}}$ with *no* drift term. This is provably wrong: applying Itô’s Lemma to check it, $Z_t = \exp\left(-\int_0^t \theta_s dW_s^{\mathbb{P}}\right)$ would satisfy $dZ_t = Z_t\left(\frac{1}{2}\theta_t^2 dt - \theta_t dW_t^{\mathbb{P}}\right)$ — a spurious drift term generated purely by exponentiating. Since $(*)$ demands *zero* drift, the exponent must carry a compensating drift term from the outset, chosen to cancel exactly what Itô’s correction introduces. Positivity of Z_t (Proposition 2.5, strict form) further forces the exponential parametrisation, since $Z_t = e^{X_t} > 0$ automatically for any process X_t .

Proposition 2.14 (Solution of the RND SDE).

$$Z_t = \exp\left(-\int_0^t \theta_s dW_s^{\mathbb{P}} - \frac{1}{2} \int_0^t \theta_s^2 ds\right).$$

Proof. Write $Z_t = e^{X_t}$ with $dX_t = a_t dt + b_t dW_t^{\mathbb{P}}$ (drift a_t , diffusion b_t both unknown). Applying Itô’s Lemma with $f(x) = e^x$, so $f_x = f_{xx} = e^x = Z_t$:

$$dZ_t = Z_t dX_t + \frac{1}{2} Z_t (dX_t)^2 = Z_t(a_t dt + b_t dW_t^{\mathbb{P}}) + \frac{1}{2} Z_t b_t^2 dt = Z_t(a_t + \frac{1}{2} b_t^2) dt + Z_t b_t dW_t^{\mathbb{P}}.$$

Matching against $(*)$, $dZ_t = -Z_t \theta_t dW_t^{\mathbb{P}} + 0 \cdot dt$:

$$\text{diffusion: } b_t = -\theta_t, \quad \text{drift: } a_t + \frac{1}{2} b_t^2 = 0 \implies a_t = -\frac{1}{2} \theta_t^2.$$

Integrating, with $X_0 = 0$ (since $Z_0 = e^{X_0} = 1$):

$$X_t = -\frac{1}{2} \int_0^t \theta_s^2 ds - \int_0^t \theta_s dW_s^{\mathbb{P}},$$

and $Z_t = e^{X_t}$ gives the claimed formula. □

Theorem 2.15 (Girsanov’s Theorem — summary). *Let θ_t satisfy Novikov’s condition $\mathbb{E}^{\mathbb{P}}\left[\exp\left(\frac{1}{2} \int_0^T \theta_s^2 ds\right)\right] < \infty$, and define Z_t as above. Then Z_t is a genuine $\mathbb{P}$ -martingale, $d\mathbb{Q}/d\mathbb{P}|_{\mathcal{F}_t} = Z_t$ defines a valid probability measure $\mathbb{Q} \sim \mathbb{P}$, and*

$$W_t^{\mathbb{Q}} = W_t^{\mathbb{P}} + \int_0^t \theta_s ds$$

is a $\mathbb{Q}$ -Brownian motion.

2.6 Verification: $\mathbb{E}^{\mathbb{P}}[Z_T] = 1$

Proposition 2.16. *For constant $\theta_t \equiv \theta$, $\mathbb{E}^{\mathbb{P}}[Z_T] = 1$.*

Proof. $Z_T = \exp(-\theta W_T^{\mathbb{P}} - \frac{1}{2} \theta^2 T)$. Since $W_T^{\mathbb{P}} \sim \mathcal{N}(0, T)$, $-\theta W_T^{\mathbb{P}} \sim \mathcal{N}(0, \theta^2 T)$. Using the Gaussian MGF $\mathbb{E}[e^X] = e^{v/2}$ for $X \sim \mathcal{N}(0, v)$:

$$\mathbb{E}^{\mathbb{P}}[Z_T] = e^{-\theta^2 T/2} \cdot \mathbb{E}^{\mathbb{P}}[e^{-\theta W_T^{\mathbb{P}}}] = e^{-\theta^2 T/2} \cdot e^{\theta^2 T/2} = 1. \quad \square$$

Remark 2.17. The two exponents cancel *exactly* — this is not incidental. The $-\frac{1}{2} \theta_s^2 ds$ term in Z_t was forced by the zero-drift requirement in Proposition 2.12; this MGF computation is the same fact viewed differently: the correction exactly compensates the variance picked up by exponentiating a Gaussian, keeping the mean pinned at 1. Verified numerically via Monte Carlo in `validate.py`.

3 The Feynman–Kac Representation

3.1 Plain Feynman–Kac

Theorem 3.1 (Feynman–Kac). *Let $dX_t = \mu(X_t, t) dt + \sigma(X_t, t) dW_t$ and define*

$$u(x, t) = \mathbb{E}[g(X_T) \mid X_t = x]$$

for a payoff function g . Then u solves the terminal-value PDE

$$u_t + \mu(x, t)u_x + \frac{1}{2}\sigma(x, t)^2u_{xx} = 0, \quad u(x, T) = g(x).$$

Proof. By construction, $u(X_t, t) = \mathbb{E}[g(X_T) \mid \mathcal{F}_t]$ is a conditional expectation of a fixed terminal quantity, hence a $\mathbb{P}$ -martingale (same structural fact as Proposition 2.7). Apply Itô’s Lemma:

$$du = (u_t + \mu u_x + \frac{1}{2}\sigma^2 u_{xx}) dt + \sigma u_x dW_t.$$

A martingale has zero drift, so the dt -coefficient vanishes, giving the PDE. The terminal condition follows since $X_T = x$ trivially yields $u(x, T) = g(x)$. $\square$

3.2 Discounting: from Feynman–Kac to the Black–Scholes PDE

Plain Feynman–Kac is not yet the pricing PDE: no-arbitrage requires the *discounted* price process to be a martingale under $\mathbb{Q}$, not the price itself.

Theorem 3.2 (Discounted Feynman–Kac / Black–Scholes PDE). *Let $C(S, t)$ be an option price with $dS_t = rS_t dt + \sigma S_t dW_t^{\mathbb{Q}}$ under the risk-neutral measure. If $e^{-rt}C(S_t, t)$ is a $\mathbb{Q}$ -martingale, then*

$$C_t + rSC_S + \frac{1}{2}\sigma^2 S^2 C_{SS} - rC = 0$$

Proof. Apply the Itô product rule to $e^{-rt}C(S_t, t)$:

$$d(e^{-rt}C) = -re^{-rt}C dt + e^{-rt}dC.$$

Expand dC via Itô’s Lemma with the risk-neutral dynamics of S_t :

$$dC = (C_t + rSC_S + \frac{1}{2}\sigma^2 S^2 C_{SS}) dt + \sigma SC_S dW_t^{\mathbb{Q}}.$$

Substituting,

$$d(e^{-rt}C) = e^{-rt}[-rC + C_t + rSC_S + \frac{1}{2}\sigma^2 S^2 C_{SS}] dt + e^{-rt}\sigma SC_S dW_t^{\mathbb{Q}}.$$

The martingale property forces the drift to vanish, giving the boxed PDE. $\square$

Proposition 3.3 (Consistency check). *Let $v(x, t) = \mathbb{E}^{\mathbb{Q}}[g(S_T) \mid S_t = x]$, which by Theorem 3.1 solves the plain (undiscounted) PDE $v_t + rxv_x + \frac{1}{2}\sigma^2 x^2 v_{xx} = 0$ with $v(x, T) = g(x)$. Then $u(x, t) := e^{-r(T-t)}v(x, t)$ solves the discounted PDE of Theorem 3.2.*

Proof. Compute $u_t = ru + e^{-r(T-t)}v_t$, $u_x = e^{-r(T-t)}v_x$, $u_{xx} = e^{-r(T-t)}v_{xx}$. Then

$$u_t + rxu_x + \frac{1}{2}\sigma^2 x^2 u_{xx} - ru = e^{-r(T-t)} \underbrace{\left[v_t + rxv_x + \frac{1}{2}\sigma^2 x^2 v_{xx} \right]}_{=0, \text{ plain FK}} = 0. \quad \square$$

Remark 3.4. This closes the loop connecting the three results of these notes: Girsanov supplies $\mathbb{Q}$ and the risk-neutral dynamics of S_t ; Feynman–Kac connects conditional expectations to PDEs; discounting converts that into the actual Black–Scholes PDE, matching the replication-argument derivation via an entirely different (probabilistic) route.

Computational Validation

The accompanying `validate.py` script numerically confirms:

- (i) the GBM log-drift correction: empirical drift of $\ln S_t$ matches $\mu - \frac{1}{2}\sigma^2$, not the naive μ ;
- (ii) the Girsanov martingale property $\mathbb{E}^{\mathbb{P}}[Z_T] = 1$ via Monte Carlo;
- (iii) the measure-change consequence $\mathbb{E}^{\mathbb{Q}}[S_T] = S_0 e^{rT}$, computed as $\mathbb{E}^{\mathbb{P}}[Z_T \cdot S_T]$ under $\mathbb{P}$.